

IoT Viva Cheat Sheet (One-Line Revision)

IoT – Network of smart devices exchanging data over internet.

Arduino IDE – Software to write and upload code to microcontrollers.

Preferences – Used to configure IDE and add ESP32 board URL.

COM Port – Communication port between PC and microcontroller.

digitalWrite() – Sends HIGH/LOW output signal.

digitalRead() – Reads HIGH/LOW input signal.

analogRead() – Reads analog input (0–1023 Arduino, 0–4095 ESP32).

analogWrite() – Generates PWM signal (0–255).

PWM – Controls power using ON/OFF pulses.

Duty Cycle – Percentage of ON time in signal.

ADC Resolution – Number of digital values for analog input.

INPUT_PULLUP – Internal resistor makes default HIGH, pressed LOW.

delay() – Pauses execution in milliseconds.

map() – Converts value from one range to another.

Push Button – Digital input switch.

LED – Emits light when current flows.

LDR – Resistance decreases in light, increases in dark.

PIR Sensor – Detects motion using infrared.

DHT22 – Digital temperature and humidity sensor.

Ultrasonic Sensor – Measures distance using sound waves.

pulseIn() – Measures duration of HIGH signal.

Servo Motor – Rotates 0°–180° using PWM.

Potentiometer – Variable resistor for analog input.

ESP32 – Microcontroller with WiFi/Bluetooth handling all signals.

Final: IoT uses sensors and actuators controlled by ESP32 for automation.